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Composed Image Retrieval (CIR)

$\mathcal{I}$: space of all images

$\mathcal{T}$: space of all text descriptions

in CIR we are given:

1. reference image $I_r \in \mathcal{I}$
2. modification text $T_m \in \mathcal{T}$
3. gallery $G \subset \mathcal{I}$

goal : find optimal target image $I_t^* \in G$ that reflects reference image altered by modification text i.e.

$$I_t^* = \underset{I \in G}{\arg\max}\; S\left[\phi_{comp}(I_r, T_m), \phi_{img}(I)\right]$$

where S : similarity score (usually cosine sim.)

$\phi_{comp} : I \times T \to R^d$

a composition func^n (NN) that fuses the reference image & modification text into a d-dimensional multimodal query vector.

$\phi_{img} : I \to R^d$ image encoder that maps candidate images from the gallery into the same d-dimensional space.

Supervised CIR

* triplet based training

* $D_{train} = \{(I_r^i, T_m^i, I_t^i)\}_{i=1}^{N}$

* learn NN params θ^* s.t

$$\theta^* = \arg\min_{\theta} \sum_{i=1}^{N} L\left[\phi_{comp}(I_r^i, T_m^i; \theta), \phi_{img}(I_t^i)\right] \quad \text{--- (1)}$$

Zero-shot CIR

* no triplet data

* popular method:

$I_r \xrightarrow{P}$ pseudo words $+$ T_m $\Big\} \rightarrow \phi_{text}$

* project I_r into text space

$$P(I_r) = M\left[\phi_{img}(I_r)\right] \in \mathbb{R}^{k \times d}$$

where ϕ_{img} : frozen pre-trained img encoder

M : MLP or transformer
projects visual features into text embedding space

k : # of pseudo-word tokens

d : dimensionality of text embed space.

* then tokenize T_m into $(t_1, t_2, \ldots, t_L)$

* Concatenate projected I_r & T_m and generate query vector

Query vector $= \phi_{text}\left[\left(\rho(I_r), t_1, t_2, \ldots, t_L \right) \right]$

where ϕ_{text} : pre-trained text encoder (text transformer from CLIP)

$$\phi_{text} : T \to \mathbb{R}^d$$

brief explanation of ϕ_{text} :

if T : input text

$$T \xrightarrow{\text{tokenize}} t_1, t_2, \ldots t_L$$

$$E_0 = [t_{cls}, t_1, \ldots t_L] + E_{pos}$$

$$E_k = \text{TransformerLayer}(E_{k-1})$$

for $k = 1, \ldots, K$

$$\phi_{text}(T) = E_K^{EOS} \cdot W_{proj} \in \mathbb{R}^d$$

* Consider eq^n (1)

in ZS-CIR,

$$\phi_{comp}(I_r, T_m) \approx \phi_{text}\left[(P(I_r), T_m)\right]$$

let $\quad \underline{q} = \phi_{text}\left[(P(I_r), T_m)\right] \in \mathbb{R}^d$

let $\quad \underline{v_c} = \phi_{img}(I_c) \quad \in \mathbb{R}^d$

where I_c is candidate img from G

* $\underline{q}$ & $\underline{v_c}$ are considered aligned due to CLIP pre-training

* Compute cosine similarity

$$S(\underline{q}, \underline{v_c}) = \frac{\underline{q} \cdot \underline{v_c}}{\|\underline{q}\| \, \|\underline{v_c}\|}$$

* Retrieval

$$I_t^* = \arg\max_{I_c \in G} \left[\frac{\xi \cdot \phi_{img}(I_c)}{\|\xi\| \, \|\phi_{img}(I_c)\|} \right]$$

* Contrast is generation which is solvable by large models

$$I_{out} \sim P(I \mid I_r, T_m)$$

* model is free to construct perfect target image.

1 image $\longrightarrow$ 250 - 1500 tokens

current model content window

Gemini	3.1	1 M tokens
GPT	5.4	1.05 M
Opus	4.6	200 K
Deepseek	V3	128 K

optimistic max # of images in content

$$= \frac{1\,M}{500} = 2000$$

CIR Sallary 1Yr:

fashion IQ ~ 50K

CIRCO ~ 120K

CIRR ~ 17K – 21K